

Multiple Choice (10 marks)

1. How does a buffer overflow on the stack facilitate running attacker-injected code?
 - (a) By overwriting the return address to point to the location of that code
 - (b) By writing directly to the instruction pointer register the address of the code
 - (c) By writing directly to %eax the address of the code
 - (d) By changing the name of the running executable, stored on the stack
2. The _____ was a huge marketplace of Dark Web specifically famous for selling of illegal drugs & narcotics as well as you can find a wide range of other goods for sale.
 - (a) Silk Road
 - (b) Cotton Road
 - (c) Dark Road
 - (d) Drug Road
3. Which of the following does authentication aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
4. Which of the following is not a transport layer vulnerability?
 - (a) Mishandling of undefined, poorly defined variables
 - (b) The Vulnerability that allows "fingerprinting" & other enumeration of host information
 - (c) Overloading of transport-layer mechanisms
 - (d) Unauthorized network access
5. What are the types of scanning?
 - (a) Port, network, and services
 - (b) Network, vulnerability, and port
 - (c) Passive, active, and interactive
 - (d) Server, client, and network

True / False (10 marks)

Answer True or False

6. True or False: Modern operating systems like Windows 10 are frequently targeted by buffer-overflow attacks, unlike older systems such as UNIX.
7. True or False: Buffer-overflow issues can be effectively mitigated through comprehensive memory checks alone.
8. True or False: Public key encryption simplifies the key exchange process, allowing secure communication without prior sharing of a symmetric key.
9. True or False: A packet filter firewall functions at the application or transport layer, analyzing data for specific application protocols.
10. True or False: A digital signature is created using a public-key system to ensure authenticity and integrity of the signed data.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the difference between two consecutive numbers in a given list.
12. Write a python function to find the smallest prime divisor of a number.

End of Paper